Faster **prediction** of **wildfire** behaviour by physical models through application of proper orthogonal decomposition

E Guelpa, A Sciacovelli, V Verda… - … Journal of **Wildland Fire**, 2016 - CSIRO Publishing

… Reliable and computationally feasible **prediction** of **wildland fire** behaviour (eg rate of **spread**
[… A literature **review** revealed no applications of POD to **wildland fire** modelling. This paper …

☆ Save 99 Cite Cited by 22 Related articles All 6 versions

Firemap: A dynamic data-driven predictive **wildfire** modeling and visualization environment

D Crawl, J Block, K Lin, I Altintas - Procedia Computer Science, 2017 - Elsevier

… predict the **behavior** of the … **simulation**, **prediction**, and visualization of **wildfire behavior**.
One goal of WIFIRE is to provide the tools to predict a more accurate rate of a spreading **wildfire**. …

☆ Save 99 Cite Cited by 41 Related articles

Evaluating crown fire rate of **spread** predictions from physics-based models

CM Hoffman, J Canfield, RR Linn, W Mell, CH Sieg… - Fire Technology, 2016 - Springer

… a variety of **wildland fire behavior** questions, there is a need to assess **model** performance,
especially in terms of crown fire **behavior**, to identify potential **model** errors and establish …

☆ Save 99 Cite Cited by 122 Related articles All 14 versions

Estimating **wildfire behavior** and effects

FA Albini - 1976 - books.google.com

… those concerned with **wildland fire behavior** and effects. Although fire **behavior prediction** is
by … Here we **review** some **model** uses and indicate the type of output needed and the general …

☆ Save 99 Cite Cited by 1319 Related articles All 10 versions

Next day **wildfire spread**: A machine learning dataset to predict **wildfire** spreading from remote-sensing data

F Huot, RL Hu, N Goyal, T Sankar… - IEEE Transactions on …, 2022 - ieeexplore.ieee.org

… Day **Wildfire Spread**” dataset, we train a deep learning **model** for fire **spread prediction**. Given
a … Lampin, “A **review** of the main driving factors of **forest fire** ignition over Europe,” Environ. …

☆ Save 99 Cite Cited by 129 Related articles All 5 versions

Using cellular automata to simulate **wildfire propagation** and to assist in fire management

JG Freire, CC DaCamara - Natural hazards and earth system …, 2019 - nhess.copernicus.org

… As mentioned in the introduction, we apply a CA **model** to a large and well-documented
wildfire that occurred in July 2012 in the Tavira and São Brás de Alportel municipalities, located …

☆ Save 99 Cite Cited by 102 Related articles All 15 versions »

Parameter flexible **wildfire prediction** using machine learning techniques: Forward and inverse modelling

S Cheng, Y Jin, SP Harrison, C Quilodrán-Casas… - Remote Sensing, 2022 - mdpi.com

… observations; (ii) the forward fire **prediction model** once combined with ML approaches; (iii) …
propagation using estimated/adjusted parameters. After ROM, we aim to learn the **dynamics** …

☆ Save 99 Cite Cited by 57 Related articles All 14 versions »

Empirical-based models for predicting head-fire rate of **spread** in Australian fuel types

MG Cruz, JS Gould, ME Alexander, AL Sullivan… - Australian …, 2015 - Taylor & Francis

… of each **model** of head-fire rate of **spread** and their … -fire rate of **spread** described in this
review are listed in Table … In presenting the various **bushfire** rate of **spread** models used in …

☆ Save 99 Cite Cited by 119 Related articles All 5 versions

A two-dimensional reaction-advection-diffusion **model** of the **spread** of fire in wildlands

P Grasso, M Innocente - Advances in **Forest Fire** …, 2018 - pureportal.coventry.ac.uk

… The aim of this research is to **model** the **spread** of **forest fire** efficiently in order to support the design of innovative fire management and suppression technologies and strategies, while …

☆ Save 99 Cite Cited by 16 Related articles All 2 versions »

Stochastic modelling of **wildfire spread**

S Masoudian - 2024 - search.proquest.com

… **review** - This chapter provides an overview of the mathematical modeling of **wildfire spread** … and another generated using the **Wildfire** Dynamic Simulator (wfds) **simulation**. The aim is to …

☆ Save 99 Cite Cited by 1 Related articles All 2 versions

Predicting fire **spread** and behaviour on the fireline. **Wildfire** analyst pocket: A mobile app for **wildland fire prediction**

S Monedero, J Ramirez, A Cardil - Ecological Modelling, 2019 - Elsevier

… at providing real-time analysis of **wildfire** behaviour and **spread** in order to support real-… of **spread**, flame length, fireline intensity, crown fire potential, among others) for each **simulation** …

☆ Save 99 Cite Cited by 72 Related articles All 6 versions

Assessing WildfireGPT: a comparative analysis of AI models for quantitative **wildfire spread prediction**

M Ramesh, Z Sun, Y Li, L Zhang, SK Annam, H Fang… - Natural Hazards, 2025 - Springer

… for **wildfire** forecasting, focusing on its limitations in quantitative predicting Fire Radiative Power (FRP) **spread** and comparing its performance with a specialized predictive **model** based …

☆ Save 99 Cite Cited by 2 Related articles All 3 versions

Developing the US **wildland fire** decision support system

EK Noonan-Wright, TS Opperman… - Journal of …, 2011 - Wiley Online Library

… WFDSS accesses national weather data and forecasts, fire **behavior prediction**, economic … The objective of this paper is to **review** the US **Wildland Fire** Decision Support System. We …

☆ Save 99 Cite Cited by 179 Related articles All 8 versions

Global pyrogeography: the current and future distribution of **wildfire**

MA Krawchuk, MA Moritz, MA Parisien, J Van Dorn… - PloS one, 2009 - journals.plos.org

… fire in research on global vegetation-climate change **dynamics** and conservation planning. … we built our **wildfire** occurrence models from the GFDL CM2.1 historical **simulation** data, from …

☆ Save 99 Cite Cited by 1133 Related articles All 25 versions »

Review of formal methodologies for wind–slope correction of **wildfire** rate of **spread**

JJ Sharples - International Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… to compare the quality of **prediction** of various fire **spread** models incorporating wind and … a detailed **review** of the formal wind–slope correction methods used in a suite of fire **spread** …

☆ Save 99 Cite Cited by 82 Related articles All 11 versions

Wildfirespreadts: A dataset of multi-modal time series for **wildfire spread prediction**

S Gerard, Y Zhao, J Sullivan - Advances in Neural …, 2023 - proceedings.neurips.cc

… of a **wildfire**. The only dataset we are aware of that represents **wildfire spread** predictions at a … In section 3, we describe related work in the field of **wildfire spread prediction**. We apply …

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Generative AI for Predicting 2D and 3D **Wildfire Spread**: Beyond Physics-Based Models and Traditional Deep Learning

H Xu, S Zlatanova, R Liang, I Canbulat - arXiv preprint arXiv:2506.02485, 2025 - arxiv.org

… apply generative AI models to **wildfire spread prediction**, discussing the underlying rationale … **review** of studies applying generative AI to **wildfire prediction**, highlighting their strengths for …

☆ Save 99 Cite Related articles All 2 versions »

Prediction of **forest fire spread** rate using UAV images and an LSTM **model** considering the interaction between fire and wind

X Li, H Gao, M Zhang, S Zhang, Z Gao, J Liu, S Sun… - Remote Sensing, 2021 - mdpi.com

… new neural network **model** to predict the **spread** rate of the **forest fire**. In … **forest fire spread** speed can be predicted more accurately if the wind speed is considered in the **prediction model**…

☆ Save 99 Cite Cited by 34 Related articles All 8 versions »

FireCast: Leveraging Deep Learning to Predict **Wildfire Spread**.

D Radke, A Hessler, D Ellsworth - IJCAI, 2019 - ijcai.org

… areas surrounding a burning **wildfire** have high-risk of near-future **wildfire spread**, based on historical … to a random **prediction model** and a commonly used **wildfire spread model**, Farsite, …

☆ Save 99 Cite Cited by 159 Related articles All 4 versions »

Tamm **review**: The effects of prescribed fire on **wildfire** regimes and impacts: A framework for comparison

ME Hunter, MD Robles - Forest Ecology and Management, 2020 - Elsevier

… **wildfire** regimes, and their associated ecological and societal effects, focusing specifically on **simulation** modeling studies that consider **wildfire** … ecosystem carbon **dynamics** are much …

☆ Save 99 Cite Cited by 105 Related articles All 10 versions

Large **forest fire spread prediction**: data and computational science

T Artés, A Cortés, T Margalef - Procedia Computer Science, 2016 - Elsevier

… Therefore, a **model** must be introduced to estimate the wind along the terrain. … the **prediction model** works when coupling both models wind field **model** and **forest fire propagation model**. …

☆ Save 99 Cite Cited by 15 Related articles All 6 versions

Modeling **forest fire spread** using machine learning-based cellular automata in a GIS environment

Y Xu, D Li, H Ma, R Lin, F Zhang - Forests, 2022 - mdpi.com

… In this paper, we propose a machine learning-based **forest fire spread model** in which the traditional cellular automata framework is combined with LSSVM (least squares support vector …

☆ Save 99 Cite Cited by 30 Related articles All 6 versions »

Atmospheric turbulence and wildland fires: a **review**

WE Heilman - International journal of **wildland fire**, 2023 - CSIRO Publishing

… Palmer also presented a conceptual **model** to explain a spiralling turbulent twin-vortex … high-resolution simulations of ABL turbulence **dynamics** during **wildland fire** events are needed to …

☆ Save 99 Cite Cited by 12 Related articles All 8 versions

A **simulation** of probabilistic **wildfire** risk components for the continental United States

MA Finney, CW McHugh, IC Grenfell, KL Riley… - … Research and Risk …, 2011 - Springer

… To make the fire growth **simulation** efficient, fire **behavior** for the entire landscape was pre-processed for all combinations of moisture (three percentile categories: 80th, 90th, and 97th …

☆ Save 99 Cite Cited by 487 Related articles All 23 versions

Advancements in **forest fire** prevention: A comprehensive **survey**

F Carta, C Zidda, M Putzu, D Loru, M Anedda, D Giusto - Sensors, 2023 - mdpi.com

… [118] introduced a **wildfire spread model** implementing an … fire **simulation** models were conducted on a real **wildfire** in Getty, California, USA. The results demonstrate that the IGN **model** …

☆ Save 99 Cite Cited by 123 Related articles All 12 versions »

Adaptive **wildfire spread prediction** for complex terrain: modeling the effectiveness of sprinkler systems

J Kim, J Ahn, J Kang - Fire Ecology, 2024 - Springer

… of a comprehensive CFD **model** in mitigating **wildfire spread** using sprinkler systems as an … in **wildfire** within 10 h of ignition, significant influence of wind speed on **spread** patterns, and a …

☆ Save 99 Cite Cited by 5 Related articles All 8 versions

Comparing Accuracy of **Wildfire Spread Prediction** Models under Different Data Deficiency Conditions

J Zhou, W Jiang, F Wang, Y Qiao, Q Meng - Fire, 2024 - mdpi.com

… the **prediction** accuracy of existing **wildfire spread** models and … **behavior** yet identified. This study aims to simulate the impact of data scarcity on the accuracy of **wildfire spread prediction** …

☆ Save 99 Cite Cited by 1 Related articles All 4 versions »

… data-driven simulations of **wildfire spread**–Part I: Reduced-cost Ensemble Kalman Filter based on a Polynomial Chaos surrogate **model** for parameter estimation

MC Rochoux, S Ricci, D Lucor… - Natural Hazards and …, 2014 - nhess.copernicus.org

… fire **spread model**; FIREFLY could readily be replaced by any other front-tracking **wildfire spread** simulator, for instance FARSITE, FOREFIRE, PROMETHEUS or PHOENIX RapidFire. …

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Simulating **wildfire spread** based on continuous time series remote sensing images and cellular automata

H Zhuang, N Liu, X Xie, X Xu, M Li… - … of **Wildland Fire**, 2025 - CSIRO Publishing

… CA **model** is proposed to accurately simulate **wildfire spread** … a **simulation** system for predicting **wildfire propagation** that … to this paper during the peer **review** process. All authors hereby …

☆ Save 99 Cite Related articles All 2 versions

Review of **wildfire** management techniques—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… in consecutive frames over time for remote monitoring of **wildfire spread**. In [45], a … **forest fire** with low consumption of energy. The fire **spread model** in [46] can produce fire **propagation** …

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Integrating **wildfire spread** and evacuation times to design safe triggers: Application to two rural communities using PERIL **model**

H Mitchell, S Gwynne, E Ronchi, N Kalogeropoulos… - Safety science, 2023 - Elsevier

… the **wildfire** indicate it is time to evacuate. Triggers are often arbitrarily defined via identifiable landmarks, do not include a safety factor, and do not account for fire **spread** … of **wildfire** and …

☆ Save 99 Cite Cited by 25 Related articles All 7 versions

The importance of fire–atmosphere coupling and boundary-layer turbulence to **wildfire spread**

R Sun, SK Krueger, MA Jenkins… - … of **Wildland Fire**, 2009 - CSIRO Publishing

… The major source of uncertainty in **wildfire behavior prediction** is the transient **behavior** of **wildfire** due … The coupled **wildfire**–atmosphere **model** used in the present study is limited by the …

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Review of **wildfire** modeling considering effects on land surfaces

D Or, E Furtak-Cole, M Berli, R Shillito… - Earth-Science …, 2023 - Elsevier

… **review** are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances in modeling of **wildfire dynamics** and rates of **spread** … large eddy **simulation** models. We …

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Review of **Wildfire** Detection, Fighting, and Technologies: Future Prospects and Insights

NM Negash, L Sun, C Fan, D Shi, F Wang - AIAA AVIATION FORUM …, 2025 - arc.aiaa.org

… management tasks depicted in the framework include fire detection, **wildfire** segmentation, **spread prediction**, and risk assessment, aligning with the capabilities commonly emphasized …

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Wildland fire prevention: a **review**

H Hesseln - Current Forestry Reports, 2018 - Springer

… **behavior** and factors that will lead to enhanced **wildfire** … methods to **model** the likelihood of **wildfire** including those … spatio-temporal dimensions and **model** outputs. Their findings …

☆ Save 99 Cite Cited by 78 Related articles All 4 versions

Coupling wind **dynamics** into a DDDAS **forest fire propagation prediction** system

C Brun, T Artées, T Margalef, A Cortées - Procedia Computer Science, 2012 - Elsevier

… **prediction** quality improvements when coupling **forest fire spread model** with weather **forecast** … Rothermel fire **propagation model**[14], has been used as **forest fire spread** simulator and, …

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The distributed strategy for asynchronous observations in data-driven **wildland fire spread prediction**

M Zha, Z Wang, J Ji, J Zhu - International Journal of **Wildland Fire**, 2024 - CSIRO Publishing

… as the basic fire **propagation model**, which is widely used for **wildfire prediction**. Real terrain, fuel … the **prediction**. Fuel adjustment factors and wind parameters were randomly generated. …

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DEVS-FIRE: design and application of formal discrete event **wildfire spread** and suppression models

X Hu, Y Sun, L Ntaimo - **Simulation**, 2012 - journals.sagepub.com

… **simulation model** and uses spatial fire **behavior** input data on topography and the Canadian **forest fire behavior prediction** (… to modeling the **spread** of **wildland fire**: A **review**. Prog Phys …

☆ Save 99 Cite Cited by 84 Related articles All 5 versions

Wildfire spread modeling with aerial image processing

Q Huang, A Razi, F Afghah… - 2020 IEEE 21st …, 2020 - ieeexplore.ieee.org

… We evaluate the fire **propagation model** by comparing the results with the … modified and complements the Rothermel’s **model**, we first **review** this **model**, and then mention how we …

☆ Save 99 Cite Cited by 30 Related articles All 3 versions

Wildfire Models–Concise **Review**

KRA Pillai, YEAD Sultan - International Journal of Innovative …, 2023 - researchgate.net

… of **wildfire** modeling is to understand and **forecast** fire **behavior** using numerical **simulation** … Past models of **wildfire spread** have included basic ellipses, egg- and fan-shaped models, …

☆ Save 99 Cite Cited by 2 Related articles »

Wildfire behaviour and danger ratings

E Planas, E Pastor - Fire phenomena and the Earth system: an …, 2013 - Wiley Online Library

… the evolution and **dynamics** of a **wildfire**, namely the way fire … **prediction** systems (eg, see Chapter 3 for a historical **review**). … and referenced **wildland fire** behaviour **model** ever developed…

☆ Save 99 Cite Cited by 13 Related articles All 2 versions

Towards data-driven operational **wildfire spread** modeling: A report of the NSF-funded WIFIRE workshop

M Gollner, A Trouve, I Altintas, J Block, R de Callafon… - 2015 - drum.lib.umd.edu

… -empirical rate-of-**spread wildfire model**. Simulations with ABL … -**spread wildfire model** operates on a separate surface **model** … , our **review** will restrict itself to experiments focused on fire …

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Generative algorithms for fusion of physics-based **wildfire spread** models with satellite data for initializing **wildfire** forecasts

B Shaddy, D Ray, A Farguell, V Calaza… - … Intelligence for the …, 2024 - journals.ametsoc.org

… Mandel, 2018: Assimilation of fire perimeters and satellite detections by minimization of the residual in a fire **spread model**. Computational Science–ICCS 2018, Y. Shi et al., Eds., …

☆ Save 99 Cite Cited by 8 Related articles All 5 versions

Towards predictive **simulation** of **wildfire spread** at regional scale using ensemble-based data assimilation to correct the fire front position

MC Rochoux, C Emery, S Ricci, B Cuenot… - Fire Safety …, 2014 - publications.iafss.org

… The prototype **simulation** capability is then evaluated in a validation test corresponding to a … providing an optimized **forecast** of the **wildfire behavior**. KEYWORDS: **wildfire**, fire modeling, …

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Operational **Forest-Fire spread** forecasting using the WRF-SFIRE **model**

MP Kale, SS Meher, M Chavan, V Kumar, MA Sultan… - Remote Sensing, 2024 - mdpi.com

… -source WRF-SFIRE **model** has been used to carry out surface **forest fire spread** forecasting in the … Global **forecast** system (GFS)-based hourly forecasted weather **model** data obtained …

☆ Save 99 Cite Cited by 6 Related articles All 5 versions »

Adaptive **forest fire spread simulation** algorithm based on cellular automata

L Sun, C Xu, Y He, Y Zhao, Y Xu, X Rui, H Xu - Forests, 2021 - mdpi.com

… automata with a fixed time step to simulate **forest fire spread** may be limited in its ability to … with an existing **forest fire model** to construct an improved **forest fire spread model**, which …

☆ Save 99 Cite Cited by 32 Related articles All 6 versions »

Attention-based **wildland fire spread** modeling using fire-tracking satellite observations

Y Zou, M Sadeghi, Y Liu, A Puchko, S Le, Y Chen… - Fire, 2023 - mdpi.com

… Physics-based fire models, as outlined in the referenced **review** paper [13], are based on … We used **model** input settings in our CNN fire **spread** models that were similar to the …

☆ Save 99 Cite Cited by 10 Related articles All 9 versions »

Fifty years of progress in **wildland fire** modelling: from empirical to fully physical CFD models

D Morvan, G Accary, S Meradji… - Comptes …, 2022 - comptes-rendus.academie-sciences …

… The aim of this short **review** is to present the progress made in **wildland fire** modelling … **model** [18–20]. For operational applications, requiring the **simulation** of **wildfire propagation** at a …

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Comparison of Different Models to Simulate **Forest Fire Spread**: A Case Study

J Ning, H Liu, W Yu, J Deng, L Sun, G Yang, M Wang… - Forests, 2024 - mdpi.com

… **forest fire spread model** … **model** to predict **forest fire spread** by using FIRER in a specific forest ecosystem. The ecological significance of FIRER is that by using it for fire **spread prediction**…

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Development of a **model** system to predict **wildfire** behaviour in pine plantations

MG Cruz, ME Alexander, PAM Fernandes - Australian Forestry, 2008 - Taylor & Francis

… occasional sustained crown fire **propagation** in denser stands. … 7 **wildfire** propagated mostly as a surface fire with periodic … of a **model** system aimed at predicting the rate of **spread** and …

Save Cite Cited by 98 Related articles All 14 versions

Navigating the evolving landscape of **wildfire** management: A systematic **review** of decision support tools

T O'Mara, AS Meador, M Colavito, A Waltz… - Trees, Forests and …, 2024 - Elsevier

… to explore the landscape of decision support tools (DSTs) for **wildfire** management, … **review** led us to group decision support tools into categories for all DSTs, such as, Fire **Behavior** …

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Analysis of **wildfire** visualization systems for research and training: are they up for the challenge of the current state of wildfires?

CAT Cortes, S Thurow, A Ong… - IEEE transactions on …, 2023 - ieeexplore.ieee.org

… , this **survey** reviews various studies that use immersive and non-immersive data visualization techniques to depict **wildfire behavior** and … conditions on **simulation** of **wildfire spread**,” …

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Studying **wildfire behavior** using FIRETEC

R Linn, J Reisner, JJ Colman… - … journal of **wildland fire**, 2002 - CSIRO Publishing

… capacity for a physics-based **model** capable of describing **wildfire behavior** in a wide variety of circumstances. We believe that a three-dimensional **wildfire model** of this type could help …

☆ Save 99 Cite Cited by 563 Related articles All 5 versions

Toward a comprehensive **wildfire spread model**

RO Weber - International Journal of **Wildland Fire**, 1991 - CSIRO Publishing

… It is hoped that the presented ideas will stimulate discussion on the likely viability of a future comprehensive physical **model** of **wildfire spread**. This author believes the simple models …

☆ Save 99 Cite Cited by 101 Related articles All 3 versions

Role of flow topology in wind-driven **wildfire propagation**

S Viknesh, A Tohidi, F Afghah, R Stoll, A Arzani - Physics of Fluids, 2025 - pubs.aip.org

… In a comprehensive **review**, Sullivan (2009a) detailed the development of physics-based … Gaining such fundamental insights is crucial, as it enables the **prediction** of **wildfire spread** …

☆ Save 99 Cite Cited by 1 Related articles All 4 versions

Modeling **wildfire spread** with an irregular graph network

W Jiang, F Wang, G Su, X Li, G Wang, X Zheng… - Fire, 2022 - mdpi.com

… In this paper, we propose a **wildfire spread model** with an irregular graph … **wildfire spread model** with an irregular graph network (IGN) to simulate and analyze the spatiotemporal **spread** …

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Forest fire management and fire suppression strategies: a systematic literature **review**

B Tezcan, T Eren - Natural Hazards, 2025 - Springer

… and preventing the **spread** of fires. The combination of **simulation** and optimization techniques … This study provides a comprehensive **review** of mathematical models for **forest fire** …

☆ Save 99 Cite Cited by 6 Related articles All 6 versions

Forecasting daily **wildfire** activity using poisson regression

CA Graff, SR Coffield, Y Chen… - … on Geoscience and …, 2020 - ieeexplore.ieee.org

… We find that an intercept-only **prediction model** can capture this expected **behavior** and that predicting this overall mean trend alone is sufficient to improve on the predictive accuracy of …

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Data-driven **wildfire spread** modeling of european wildfires using a spatiotemporal graph neural network

M Rösch, M Nolde, T Ullmann, T Riedlinger - Fire, 2024 - mdpi.com

… yet applied STGNNs to **model** the daily **wildfire spread** on a … DL-based **wildfire spread model** to predict the daily **spread** of … uses a STGNN to **model** the **wildfire spread** for the first time. …

☆ Save 99 Cite Cited by 13 Related articles All 6 versions »

Climate change impact on future **wildfire** danger and activity in southern Europe: a **review**

J Dupuy, H Fargeon, N Martin-StPaul, F Pimont… - Annals of Forest …, 2020 - Springer

… moisture, which in turn conditions fire **spread** in the fire module. Both LPJ-… **spread** in SPITFIRE, while the maximum daily Nesterov index is used to calibrate the annual burnt area **model** …

☆ Save 99 Cite Cited by 385 Related articles All 20 versions

Review of vortices in **wildland fire**

JM Forthofer, SL Goodrick - Journal of Combustion, 2011 - Wiley Online Library

… fire **behavior** and safety concerns. In this paper, the current state of knowledge of the interaction of **wildland fire** … In this study a computer **model** of the atmospheric boundary layer was …

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Wildfire likelihood's elements: a literature **review**

M Mhawej, G Faour, J Adjizian-Gerard - Challenges, 2015 - mdpi.com

… that affect the ignition of the **wildfire**, its initial **spread** and its free **propagation**). The first set of … generates various positive outcomes, particularly in terms of **prediction**, prevention and …

☆ Save 99 Cite Cited by 47 Related articles All 10 versions »

Guidance on spatial **wildland fire** analysis: models, tools, and techniques

RD Stratton - 2006 - books.google.com

… If the **model** has not been properly calibrated to local fires, how … , when fire **behavior** outputs have been critiqued and the **model** … Definition of a fire **behavior model** evaluation protocol: a …

☆ Save 99 Cite Cited by 131 Related articles All 6 versions

Review and perspectives of digital twin systems for **wildland fire** management

Y Li, T Zhang, Y Ding, R Wadhwani… - Journal of Forestry …, 2024 - Springer

… **wildfire** prevention, monitoring, disaster response, and post-fire recovery. This **review** examines the potential utility of Digital Twin in **wildfire** … operation **model** to **model** the **wildfire spread** …

☆ Save 99 Cite Cited by 11 Related articles All 7 versions

Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… An example **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is shown in Figure 2 with s c = 0.5 and s w = 0.5 in Equation (8). These constants …

☆ Save 99 Cite Cited by 19 Related articles All 2 versions »

Wildfire autonomous response and **prediction** using cellular automata (WARP-CA)

A Ramadan - arXiv preprint arXiv:2407.02613, 2024 - arxiv.org

… realism in simulating **wildfire dynamics** [8, 16, 28]. Our literature **review** uncovered a lack of … generating heterogeneous terrain in **forest fire dynamics simulation**. The mathematical basis …

☆ Save 99 Cite Cited by 1 Related articles All 2 versions »

Methods for **Wildfire Spread Prediction** and Their Integration with Remote Sensing Data

SA Khvostikov, SA Bartalev - Inf. Technol. Remote Sens. Earth …, 2019 - academia.edu

… short **review** article presents description of a field of **wildfire** … just-in-time **wildfire** danger **prediction** system. Also in last 10 years … **wildfire** models, which leads to better estimation of **wildfire** …

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Convolutional LSTM neural networks for modeling **wildland fire dynamics**

J Burge, M Bonanni, M Ihme, L Hu - arXiv preprint arXiv:2012.06679, 2020 - arxiv.org

… Here, we seek to use ML alone to explicitly **model wildland fire spread dynamics** based solely on data. In effect, ML in this context represents the development of an empirical …

☆ Save 99 Cite Cited by 33 Related articles All 3 versions »

Wildfire propagation simulation tool using cellular automata and GIS

W Velasquez, A Munoz-Arcentales… - 2019 International …, 2019 - ieeexplore.ieee.org

… **Wildfire Propagation Simulation** tool able to present graphically, how the fire will be propagated in case of a **wildfire**… Calkin in ”Uncertainty and risk in **Wildland fire** management: A **review**…

☆ Save 99 Cite Cited by 11 Related articles All 3 versions

Challenges and opportunities in remote sensing-based fuel load estimation for **wildfire behavior** and management: A comprehensive **review**

A Abdollahi, M Yebra - Remote Sensing, 2025 - opus.lib.uts.edu.au

… Most fire **behavior prediction** models, including BEHAVE, FlamMap, and FARSITE require a … and **spread** of surface fires [39,46]. Dead fuels play a pivotal role in **wildfire prediction**, …

☆ Save 99 Cite Cited by 8 Related articles All 7 versions »

Wildfire mitigation plans in power systems: A literature **review**
DAZ Vazquez, F Qiu, N Fan… - IEEE Transactions on …, 2022 - ieeexplore.ieee.org
… development of **wildfire spread** simulations crucial. Energy companies use **wildfire spread simulation** in … The **wildfire spread simulation** results are used as inputs to define the risk event …
☆ Save 99 Cite Cited by 78 Related articles All 4 versions

Near real-time extracting **wildfire spread** rate from Himawari-8 satellite data
X Liu, B He, X Quan, M Yebra, S Qiu, C Yin, Z Liao… - Remote Sensing, 2018 - mdpi.com
… These methods can be divided into two categories: traditional (eg, laboratory or field observation, fire **spread model** estimation) and remotely sensing based methods (eg, extraction …
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Wildfire danger **prediction** and understanding with deep learning
S Kondylatos, I Prapas, M Ronco… - Geophysical …, 2022 - Wiley Online Library
… are of pivotal importance in modeling **wildfire dynamics**. To test the generalization robustness, … the occurrence and **spread** of wildfires and examine their temporal contribution decisions. …
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Forest fire spread simulation algorithm based on cellular automata
X Rui, S Hui, X Yu, G Zhang, B Wu - Natural hazards, 2018 - Springer
… of time step on the **simulation** of **forest fire spread**, we set the **forest fire spread** speed to a constant. Taking a fire point as the initial fire source, the **forest fire** should **spread** into outward …
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Turbulence and fire-spotting effects into **wild-land fire** simulators
I Kaur, A Mentrelli, F Bosseur, JB Filippi… - … and Numerical **Simulation**, 2016 - Elsevier
… method when applied to **wild-land fire propagation**. Simple … **wildfire propagation model** qualitatively improves its performance (eg, reproducing flank and back fire, increase in fire **spread** …
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Dynamic data driven application system for **wildfire spread simulation**
F Gu - 2010 - search.proquest.com
… of **wildfire spread** simulations, we apply the concept of DDDAS to **wildfire spread simulation** by dynamically assimilating sensor data from real wildfires into the **simulation model**. …
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Review of **wildfire** detection using social media
V Slavkovikj, S Verstockt, S Van Hoecke… - Fire safety journal, 2014 - Elsevier
… in the context of **wildfire propagation**. The authors evaluated … **model** map and used as a ground truth of the **model forecast**. … oil **propagation model** by correlating the **model** predictions …
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Efficient **simulation** of **wildfire spread** on an irregular grid
P Johnston, J Kelso, GJ Milne - … Journal of **Wildland Fire**, 2008 - CSIRO Publishing
… **simulation**, which is a much more efficient computational method than conventional **wildfire simulation** … to parts of the fire whose **dynamics** change infrequently, namely those areas of a …
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Predicting **wildland fire behavior** and emissions using a fine-scale physical **model**
B Porterie, JL Consalvi, A Kaiss… - Numerical Heat Transfer …, 2005 - Taylor & Francis
… The main purpose of this work is twofold: to describe fire **behavior** and to … **model**. Small-scale models, such as that presented in this article, may be used to quantify the impact of **wildfire** …
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A **survey** of machine learning algorithms based forest fires **prediction** and detection systems

F Abid - Fire technology, 2021 - Springer

… the **behavior** of the **forest fire**, ie, forecasting the fire **spread** evolution. The **forest fire behavior prediction** … In this study a **model** for **wildfire** ignition probabilities **prediction** was proposed. A …

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Advancements in **Wildfire Prediction** and Detection: A Systematic **Review**

V Kolluru, AN Chintakunta, Y Nuthakki, S Koganti - 2022 - papers.ssrn.com

… One big issue is the high degree of endliness of the factors that play a role in the **behavior** of fire or **wildfire**. Some factors influencing forest fires include weather conditions, physical …

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Iterated local search for the placement of **wildland fire** suppression resources

AB Mendes, FP e Alvelos - European Journal of Operational Research, 2023 - Elsevier

… **propagation** times between adjacent nodes and use the minimum travel time principle to **model** the fire **propagation** … We also provide a categorised literature **review** on fire suppression …

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A global fuel characteristic **model** and dataset for **wildfire prediction**

JR McNorton, F Di Giuseppe - Biogeosciences, 2024 - bg.copernicus.org

… The further development of a full fire risk and fire **spread model** requires consideration of ignition and **spread** based on meteorology and orography. Future modelling applications could …

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A generic, empirical-based **model** for predicting rate of fire **spread** in shrublands

WR Anderson, MG Cruz, PM Fernandes… - … of **Wildland Fire**, 2015 - CSIRO Publishing

… **prediction** errors for the **wildfire** validation datasets had no bias; this is despite the **wildfire** datasets having average rates of **spread** almost double that of the **model** development sets. …

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A **review** of approaches to estimate **wildfire** plume injection height within large-scale atmospheric chemical transport models

R Paugam, M Wooster, S Freitas… - … Chemistry and Physics, 2016 - acp.copernicus.org

… we cannot resolve the plume **dynamics**, parameterizations are … The aim of this manuscript is to **review** the different … of the Sofiev **model** against (i) the original version of the **model**, (ii) …

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Using soil moisture information to better understand and predict **wildfire** danger: a **review** of recent developments and outstanding questions

ES Krueger, MR Levi, KO Achieng… - … of **wildland fire**, 2022 - CSIRO Publishing

… **model** compared to a similar **model** without soil moisture data. … of the **wildfire** (eg rate of **spread** and fire intensity). … components describing the **wildfire** danger: **Spread** Component, …

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Using IoT and ML for **forest fire** detection, monitoring, and **prediction**: a literature **review**

M Grari, M Yandouzi, I Idrissi… - Journal of Theoretical …, 2022 - researchgate.net

… Dry vegetation fuels a **wildfire's** rapid ignition and **spread**. It … or to anticipate it, because they **spread** rapidly. Early-warning … DL to detect wildfires and their **spread** via a comprehensive …

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Fuzzy boundaries: **Simulation** and expertise in **bushfire prediction**

T Neale, D May - Social Studies of Science, 2020 - journals.sagepub.com

… ’ **prediction** (see Figure 3), is produced using a computer-based **bushfire spread** simulator. … agencies to protect themselves from ‘post-incident **review** and judgment, analysis, litigation, …

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Using spatial reinforcement learning to build forest **wildfire dynamics** models from satellite images

S Ganapathi Subramanian, M Crowley - Frontiers in ICT, 2018 - frontiersin.org

… we learn a **dynamics model** from readily available satellite image data and treating **wildfire** as an … This **review** demonstrates that while various classical AI and ML techniques have been …

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Visualization of **prediction** methods for **wildfire** modeling using CiteSpace: a bibliometric analysis

M Pan, S Zhang - Atmosphere, 2023 - mdpi.com

… It highlights the dominant areas of research, including climate change, fire **behavior**, … field of **wildfire prediction**. It facilitated a systematic **review** of the **wildfire prediction** domain through …

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Wildfire management in Canada: **Review**, challenges and opportunities

C Tymstra, BJ Stocks, X Cai, MD Flannigan - Progress in Disaster Science, 2020 - Elsevier

… We conclude with a proposed future path for **wildfire** management in Canada. To co-exist with **wildfire**, agencies in Canada must also strengthen and adjust their **wildfire** management …

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RETRACTED ARTICLE: Shapley-based interpretation of deep learning models for **wildfire spread** rate **prediction**

F Qayyum, NA Samee, M Alabdulhafith, A Aziz… - Fire Ecology, 2024 - Springer

… of **wildfire spread** rate **prediction**. … **wildfire prediction model**. As mentioned earlier, the original dataset consisted of 238 instances, which was considered inadequate for effective **model** …

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Review of the **Forest fire model** for high explosives

R Menikoff, MS Shaw - Combustion Theory and Modelling, 2008 - Taylor & Francis

… **propagation** of detonation waves in solid high explosives. Here we … **review** the assumptions of the **model**. In analogy with Whitham–Chisnell shock **dynamics**, we derive the **Forest fire** …

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Analysis of evacuation **behavior** in a **wildfire** event

T Toledo, I Marom, E Grimberg, S Bekhor - International journal of disaster …, 2018 - Elsevier

… A large-scale **wildfire** took place in Haifa on November 24, 2016. On that day, starting at around … Shortly after the fire events, a web **survey** was developed and administered in order to …

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Wildfire spread prediction using attention mechanisms in u2-net

H Xiao, Y Zhu, Y Sun, G Zhang, Z Gong - Forests, 2024 - mdpi.com

… We categorize this **wildfire spread prediction** task as a semantic segmentation problem in image segmentation, where each region is predicted to contain a fire or not based on the …

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Detecting critical nodes in forest landscape networks to reduce **wildfire spread**

D Yemshanov, N Liu, DK Thompson, MA Parisien… - PloS one, 2021 - journals.plos.org

… to plan **wildfire** fuel treatments that minimize the risk of fire **spread** in … **simulation** modeling to estimate the probability of fire **spread** … A **review** of operational research studies in **forest fire** …

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An uncertainty analysis of **wildfire** modeling

K Riley, M Thompson - Natural Hazard Uncertainty Assessment …, 2016 - Wiley Online Library

… To provide background, we **review** a simple conceptual **model** of the major factors that … We focus principally on **simulation** models that implement fire **spread** in a geospatial context …

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A control-theoretic spatio-temporal **model** for **wildfire** smoke **propagation** using UAV-based air pollutant measurements

P Ragbir, A Kaduwela, X Lan, A Watts, Z Kong - Drones, 2024 - mdpi.com

… **Wildfire** remote sensing with UAVs: A **review** from the autonomy point of view. In Proceedings of the 2020 International Conference on Unmanned Aircraft Systems (ICUAS), Athens, …

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Modeling **wildfire** incident complexity **dynamics**

MP Thompson - PloS one, 2013 - journals.plos.org

… Figure 3 presents a conceptual representation of a first-order Markov chain **model** of incident complexity **dynamics**. To simplify the presentation here only three states are used (H = high…

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Wildfire Spread Prediction Through Remote Sensing and UAV Imagery-Driven Machine Learning Models

MR Mohebbi, EW Sena, M Döller… - 2024 18th International …, 2024 - ieeexplore.ieee.org

… and mitigate **wildfire** effects using these datasets. This article presents a comprehensive **review** of how the utilization of limited UAVs image datasets affects applications in fire **trajectory** …

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